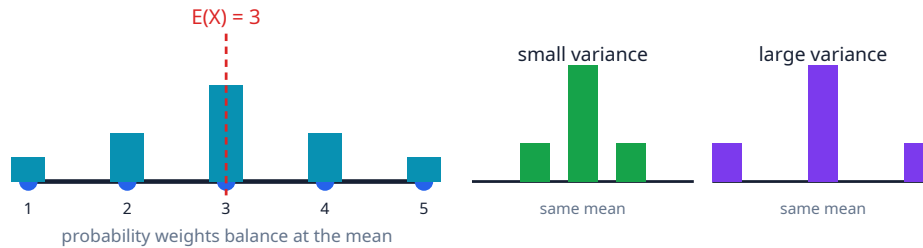


04 EXPECTATION AND VARIANCE (notes p086-p111)

1. The three parameters (p086-p087)

Mean is the balance point; variance is squared spread



Expectation

$E[h(X)] = \text{sum or integral of } h(x) \text{ times weight}$

Variance

$\text{Var}(X) = E(X^2) - [E(X)]^2$

shifts preserve variance; multiplying by a scales variance by a^2

Mean as balance point and variance as spread

MEAN μ centre / location

VARIANCE σ^2 spread

SD σ spread, in the ORIGINAL units

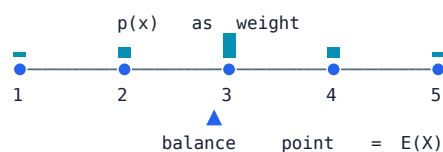
"knowledge of the numerical values of these parameters gives the researcher quick insight into the nature of the variable"

2. Expectation of a discrete rv (p088)

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

"the weighted average of the values, weighted by their probabilities"

the picture: a balance beam. values on the x-axis, probabilities as weights.
the mean is the BALANCE POINT.



3. Worked: a fair die (p089)

$$p(x) = 1/6, x = 1..6$$

$$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6)$$

$$= (1+2+3+4+5+6)/6 = 21/6 = 7/2 = 3.5$$

3.5 is not a face on the die. The mean need not be an achievable value. It is the long-run average.

4. Expectation of a function of X (p090)

$$E[h(X)] = \sum_{\{x\}} h(x) p(x) \text{ (discrete)}$$

$$E[h(X)] = \int h(x) f(x) dx \text{ (continuous)}$$

$E[h(X)] \neq h(E(X))$ in general
 you must apply h to EACH value, then weight, then sum.
 you may NOT plug the mean into h .

5. Properties of the mean (p092-p093)

THEOREM $E[c] = c$ $E[cX] = c E[X]$

proof of $E[c]$:

$$E[c] = \sum c f(x) = c \sum f(x) = c(1) = c$$

✓

proof of $E[cX]$:

$$E[cX] = \sum c x f(x) = c \sum x f(x) = c E[X]$$

✓

EXERCISE (p093):

prove $E[aX + b] = a E[X] + b$

$$E[aX+b] = \sum (ax+b)f(x) = a \sum x f(x) + b \sum f(x)$$

$$= a E[X] + b(1) = a E[X] + b$$

✓

E is LINEAR, and it is linear with NO independence needed: $E[X+Y] = E[X] + E[Y]$ ALWAYS
 $E[aX+b] = aE[X]+b$ ALWAYS

6. Worked: the p094 triple (p094) - re-asked on ETE papers

$x : -3 \ 6 \ 9$

$P(X=x): 1/6 \ 1/2 \ 1/3$

$$E(X) = (-3)(1/6) + 6(1/2) + 9(1/3)$$

$$= -0.5 + 3 + 3 = 11/2 = 5.5$$

$$E(X^2) = 9(1/6) + 36(1/2) + 81(1/3)$$

$$= 1.5 + 18 + 27 = 93/2 = 46.5$$

$$E((2X+1)^2) : \text{expand first} = E[4X^2 + 4X + 1]$$

$$= 4 E(X^2) + 4 E(X) + 1$$

$$= 4(46.5) + 4(5.5) + 1$$

$$= 186 + 22 + 1 = 209$$

THE METHOD: EXPAND, then use linearity term by term.
 Do NOT compute the 3 values of $(2x+1)^2$ by hand first
 (you can, but the algebra route is faster and safer).
 sanity: $(2x+1)^2$ for $x=-3,6,9 = 25,169,361$
 weighted: $25/6 + 169/2 + 361/3 = 4.17+84.5+120.3 = 209 \checkmark$

7. Variance (p095-p100)

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2]$$

$$\text{SD} = \sigma = \sqrt{\text{Var}(X)}$$

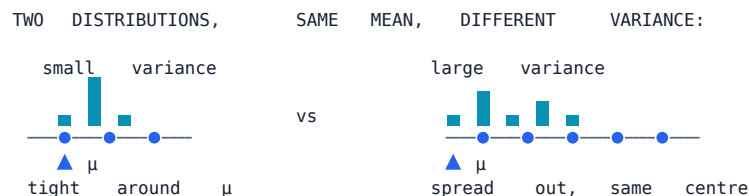
$\text{Var}(X)$ is ALWAYS non-negative (it is an expectation of a SQUARE).

the idea: the SQUARED deviation from the mean, weighted by probability.

large $\sigma^2 \iff$ more squared spread around μ

small $\sigma^2 \iff$ less squared spread around μ

Variance measures spread. It is not a universal equivalence to predictability,
 and it is unrelated to the estimator property called consistency.



8. The shortcut formula (p101-p103) - and its proof

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

PROOF:

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$= E[X^2 - 2\mu X + \mu^2]$$

expand the square

$$= E[X^2] - 2\mu E[X] + \mu^2$$

linearity

$$= E[X^2] - 2\mu\mu + \mu^2$$

since $E[X] = \mu$

$$= E[X^2] - 2\mu^2 + \mu^2$$

$$= E[X^2] - \mu^2$$

✓

THE TWO-STEP ROUTINE, ALWAYS:

1. $E(X)$

2. $E(X^2)$

3. $\text{Var} = E(X^2) - [E(X)]^2$

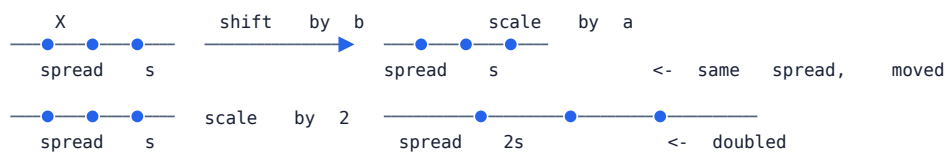
You almost never use the definition directly.

9. Variance transforms (p104-p106)

THEOREM	$\text{Var}[c] = 0$	$\text{Var}[cX] = c^2 \text{Var}[X]$
PROPOSITION $V(aX+b) = a^2\sigma^2$	$\sigma_{\{aX+b\}} = a \sigma_x$	
special cases:		
$\sigma_{\{aX\}}$	$= a \sigma_x$	scaling changes spread by $ a $
$\sigma_{\{X+b\}}$	$= \sigma_x$	SHIFTING changes nothing

WHY $|a|$ AND NOT a
 a may be negative, but a standard deviation cannot be.
 The sign of a does not affect spread, only scale.

the picture:



10. Worked: freezer dealer (p107-p108) - the 6-part chain

x : 13.5 15.9 19.1 (cubic feet)
 $p(x)$: .2 .5 .3

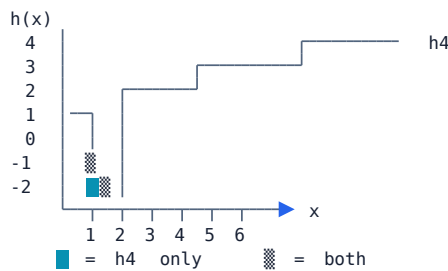
(a) $E(X) = 13.5(.2) + 15.9(.5) + 19.1(.3)$
 $= 2.70 + 7.95 + 5.73 = 16.38$
 $E(X^2) = 182.25(.2) + 252.81(.5) + 364.81(.3)$
 $= 36.45 + 126.405 + 109.443 = 272.298$
 $V(X) = 272.298 - (16.38)^2 = 272.298 - 268.3044 = 3.9936$
 (b) price = $25X - 8.5$
 $E(25X - 8.5) = 25(16.38) - 8.5 = 409.5 - 8.5 = 401$
 (c) $V(25X - 8.5) = (25)^2 V(X) = 625 \times 3.9936 = 2496$
 (the -8.5 vanishes from the variance)
 (d) actual capacity $h(X) = X - .01X^2$
 $E[h(X)] = E(X) - .01 E(X^2) = 16.38 - .01(272.298)$
 $= 16.38 - 2.723 = 13.657 \approx 13.66$

PART (d) IS THE TRAP PART.
 $E[X - .01X^2]$ is NOT $16.38 - .01(16.38)^2$.
 You must use $E(X^2)$, NOT $[E(X)]^2$.
 This is $E[h(X)] \neq h(E(X))$ in numerical form.

11. Worked: magazine orders (p109-p111) - the decision question

x : 1 2 3 4 5 6
 $p(x)$: 1/15 2/15 3/15 4/15 3/15 2/15

store pays \$1 per copy, sells at \$2, leftovers worthless.
 is it better to order 3 or 4?

net revenue as a function of demand x:					
x	:	1	2	3	4 5 6
h3(x)	:	-1	1	3	3 3 3 <- order 3: plateaus at 3
h4(x)	:	-2	0	2	4 4 4 <- order 4: plateaus at 4
p(x)	:	1/15	2/15	3/15	4/15 3/15 2/15
WHY h3 plateaus: order 3, cost 3, revenue 2 per copy SOLD (max 3 copies)					
x=1: sold 1, revenue 2, cost 3					-> -1
x=2: sold 2, revenue 4, cost 3					-> +1
x=3: sold 3, revenue 6, cost 3					-> +3
x=4: can only sell 3, revenue 6, cost 3 -> +3					(plateau: demand > stock)
$E[h_3(X)] = (-1)(1/15) + (1)(2/15) + (3)(3/15) + (3)(4/15) + (3)(3/15) + (3)(2/15)$ $= (-1+2+9+12+9+6)/15 = 37/15 = 2.4667$ $E[h_4(X)] = (-2)(1/15) + (0)(2/15) + (2)(3/15) + (4)(4/15) + (4)(3/15) + (4)(2/15)$ $= (-2+0+6+16+12+8)/15 = 40/15 = 2.6667$					
ORDER 4: higher expected revenue (2.67 vs 2.47) This is a DECISION question: express profit as h(X), take E of each option, compare.					
the two profit curves: 					

Expectation and variance done. Next: continuous random variables (p112-p147).